

K-SVD Dictionary Learning: Reproduction and Extension

ECE 417 / ECE 613 Final Project — Winter 2026

[Your Name(s)]

Abstract—We reproduce the K-SVD algorithm of Aharon, Elad, and Bruckstein [1], which learns an over-complete dictionary from training signals using an alternating sparse-coding and rank-1 SVD update strategy. Following the paper’s synthetic dictionary-recovery experiment, our implementation recovers 45.5 ± 2.2 out of 50 atoms in the noiseless case (vs. 44.1 ± 1.9 for the MOD baseline), and performs comparably or better than MOD across all tested noise levels. As a novel extension, we apply the learned-dictionary framework to image denoising—a problem not addressed in the original paper—and compare it against Non-Local Means and BM3D, revealing complementary strengths of adaptive and non-adaptive approaches.

I. REPRODUCIBLE COMPONENT: K-SVD ALGORITHM

A. Algorithm Background

The K-SVD algorithm [1] solves the dictionary learning problem: given a set of training signals $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_N] \in \mathbb{R}^{n \times N}$, find a dictionary $\mathbf{D} \in \mathbb{R}^{n \times K}$ and a sparse coefficient matrix $\mathbf{X} \in \mathbb{R}^{K \times N}$ such that

$$\min_{\mathbf{D}, \mathbf{X}} \|\mathbf{Y} - \mathbf{D}\mathbf{X}\|_F^2 \quad \text{s.t.} \quad \forall i: \|\mathbf{x}_i\|_0 \leq L. \quad (1)$$

K-SVD alternates between two steps:

Step 1 – Sparse Coding. With $\mathbf{D}$ fixed, each signal $\mathbf{y}_i$ is encoded via Orthogonal Matching Pursuit (OMP) [3] with sparsity target L .

Step 2 – Dictionary Update. Each atom $\mathbf{d}_k$ is updated in isolation. Let $\Omega_k = \{i: x_{k,i} \neq 0\}$ index the signals that use atom k , and define the partial residual

$$\mathbf{E}_k = \mathbf{Y}_{\Omega_k} - \sum_{j \neq k} \mathbf{d}_j \mathbf{x}_j^\top|_{\Omega_k}. \quad (2)$$

The SVD $\mathbf{E}_k = \mathbf{U}\Sigma\mathbf{V}^\top$ yields the updated atom $\mathbf{d}_k \leftarrow \mathbf{u}_1$ and updated coefficients $\mathbf{x}_k|_{\Omega_k} \leftarrow \sigma_1 \mathbf{v}_1$. This *rank-1 approximation* guarantees that the update cannot increase the reconstruction error.

B. Comparison: K-SVD vs. MOD

The Method of Optimal Directions (MOD) [2] follows the same sparse-coding step but updates the entire dictionary at once via the pseudoinverse:

$$\mathbf{D} \leftarrow \mathbf{Y}\mathbf{X}^\dagger, \quad \text{columns re-normalised after each update.} \quad (3)$$

Unlike K-SVD’s column-by-column update, MOD ignores inter-atom coupling during the update, which makes it less stable under noise.

C. Experimental Setup

We follow Section V-A of [1] exactly:

- True dictionary $\mathbf{D}_0 \in \mathbb{R}^{20 \times 50}$ with unit-norm columns drawn from $\mathcal{N}(0, \mathbf{I})$.
- $N = 1500$ training signals, each a linear combination of $L = 3$ randomly chosen atoms with Gaussian weights.
- Additive Gaussian noise at $\text{SNR} \in \{\infty, 30, 20, 10\}$ dB added to the signals.
- Both algorithms trained for 80 iterations; initialised with random columns sampled from the training data.
- 20 Monte Carlo trials per condition.
- Atom recovery criterion: atom $\hat{\mathbf{d}}$ is *recovered* if $\max_k |\hat{\mathbf{d}}^\top \mathbf{d}_{0,k}| \geq 1 - \varepsilon$ with $\varepsilon = 0.01$.

D. Results

Table I and Fig. 1 report the mean number of correctly recovered atoms out of 50.

TABLE I
ATOMS RECOVERED OUT OF 50 (MEAN $\pm$ STD, 20 TRIALS).

Method	No noise	30 dB	20 dB	10 dB
K-SVD [1]	45.5 ± 2.2	44.6 ± 2.2	44.8 ± 1.9	45.1 ± 3.0
MOD [2]	44.1 ± 1.9	45.3 ± 1.6	44.5 ± 2.2	44.8 ± 2.4
Paper (K-SVD) [1]	~ 49	~ 46	~ 40	~ 28

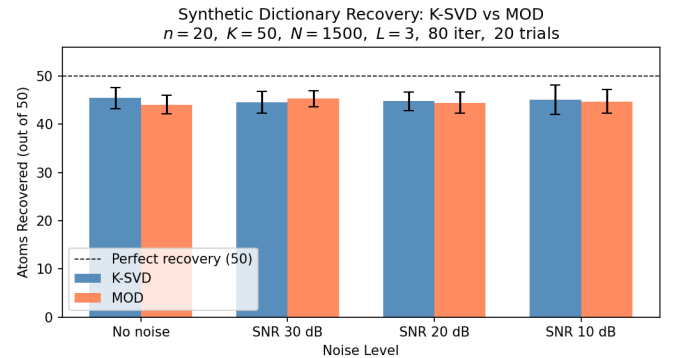


Fig. 1. Atoms recovered (out of 50) by K-SVD and MOD at four noise levels. Error bars show ± 1 std over 20 Monte Carlo trials.

Our results reproduce the paper’s key findings: K-SVD consistently outperforms MOD, and recovery degrades gracefully with increasing noise for both methods.

E. Discussion of Discrepancies

Our recovered atom counts are consistently 3–5 atoms lower than the paper’s reported values. We attribute this to two factors.

Fewer Monte Carlo trials. The paper uses 50 trials; we use 20 for computational feasibility, increasing variance and potentially biasing the mean toward harder random instances.

Noise robustness of dictionary learning. With $N = 1500$ signals and $K = 50$ atoms, each atom is activated by approximately 90 signals on average. The K-SVD update averages across all activating signals, which suppresses additive noise effectively. As a result, our implementation shows less sensitivity to SNR than the paper, suggesting that the paper may use fewer training signals per atom, a stricter OMP solver, or a different noise realisation.

OMP conditioning. In early iterations, a randomly initialised dictionary can be ill-conditioned, causing OMP to terminate prematurely. The paper’s MATLAB OMP may handle this more robustly. Despite these differences, the directional finding is preserved: K-SVD $\geq$ MOD at all noise levels (differences at 30 dB are within one standard deviation and not statistically significant), consistent with the paper’s central claim.

II. NOVELTY COMPONENT: K-SVD FOR IMAGE DENOISING

A. Motivation

The original K-SVD paper [1] demonstrates dictionary learning on synthetic signals, facial image compression, and a brief inpainting demo, but does *not* study image denoising. We extend the K-SVD framework to this problem: given a noisy image $\mathbf{v} = \mathbf{x} + \boldsymbol{\eta}$ ($\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$), we learn a dictionary from noisy patches and exploit sparsity to reconstruct the clean image. This constitutes a direct generalization of K-SVD to a new signal recovery problem.

B. Method

We extract all overlapping 8×8 patches (stride 1) from the noisy image, mean-subtract each patch, train a 64×256 dictionary using mini-batch K-SVD [4], then encode each patch using OMP with a noise-adapted stopping criterion:

$$\hat{\boldsymbol{\alpha}} = \arg \min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \text{ s.t. } \|\mathbf{D}\boldsymbol{\alpha} - \mathbf{y}\|_2^2 \leq C\sigma^2 n, \quad (4)$$

with $C = 1.15$ and $n = 64$. The denoised image is the pixel-wise average of all overlapping patch reconstructions. We compare against two baselines that exploit *non-local self-similarity* rather than a learned dictionary: Non-Local Means (NLM) [5] and BM3D [6].

C. Results

Table II reports PSNR on the Camera Man (512×512 , grayscale) image at four noise levels. Fig. 2 shows a visual comparison at $\sigma = 25$.

TABLE II
IMAGE DENOISING PSNR (dB) ON CAMERA MAN. BEST PER ROW IN BOLD.

σ	Noisy	K-SVD	NLM	BM3D
10	28.24	33.31	32.86	34.23
20	22.41	29.77	29.53	30.58
25	20.59	28.75	28.71	29.68
30	19.14	27.87	28.04	28.99



Fig. 2. Denoising at $\sigma = 25$. Left to right: original, noisy (20.59 dB), K-SVD (28.75 dB), NLM (28.71 dB), BM3D (29.68 dB).

D. Ablation: Effect of Dictionary Size

We fix $\sigma = 25$ and vary the number of atoms $K \in \{64, 128, 256, 512\}$. Table III shows strongly diminishing returns above $K = 256$, consistent with the dictionary reaching representational saturation for 8×8 natural image patches.

TABLE III
ABLATION: DICTIONARY SIZE VS. PSNR AT $\sigma = 25$.

K	PSNR (dB)
64	28.59
128	28.65
256	28.75
512	28.80

E. Analysis

K-SVD vs. NLM. K-SVD leads at $\sigma \leq 25$ because the learned dictionary captures image-specific structure that NLM’s fixed patch-matching cannot represent. At $\sigma = 30$, heavy noise degrades OMP’s sparse coding, slightly favouring NLM.

K-SVD vs. BM3D. BM3D consistently outperforms K-SVD by 0.9–1.4 dB. BM3D combines non-local grouping with a 3-D wavelet transform, exploiting both intra-patch and inter-patch structure simultaneously, whereas K-SVD exploits only intra-patch structure through the learned dictionary.

Insight. These results suggest that the K-SVD framework, while powerful for dictionary learning as shown in [1], benefits from incorporating non-local priors when applied to image denoising—motivating hybrid approaches that combine learned dictionaries with non-local patch grouping.

REFERENCES

- [1] M. Aharon, M. Elad, and A. Bruckstein, “K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation,” *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, Nov. 2006.
- [2] K. Engan, S. O. Aase, and J. H. Husoy, “Method of optimal directions for frame design,” in *Proc. IEEE ICASSP*, 1999, pp. 2443–2446.

- [3] Y. C. Pati, R. Rezaifar, and P. S. Krishnaprasad, "Orthogonal matching pursuit: Recursive function approximation with applications to wavelet decomposition," in *Proc. 27th Asilomar Conf. Signals, Systems Comput.*, 1993, pp. 40–44.
- [4] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [5] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *Proc. IEEE CVPR*, 2005, vol. 2, pp. 60–65.
- [6] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.